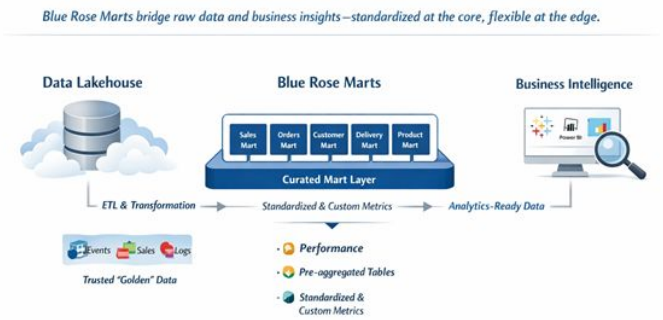
# Olist E-Commerce Data Lakehouse - Rose Marts

Preview

## From Golden Data to Business Impact: The Role of Blue Rose Marts




*Note: Alt+Click navigate buttons below.*

| Why Rose Marts Matter | What Makes Up Rose Marts | How to Leverage - BI Demo | Live Connection Steps |
|---|---|---|---|

# Why Rose Data Marts Matter?

## From Trusted 'Golden' Data Lakehouse to Actionable, Business-Ready Insights — Powered by Blue Rose Marts

**1. The Challenge (Before)**

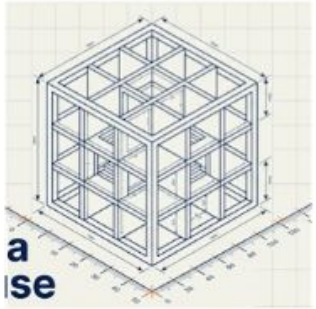
- **Data Silos**: Raw data is fragmented across sources and lacks consistency
- **High Latency**: Complex queries on raw data are slow and resource-intensive
- **Metric Drift**: Different teams report conflicting definitions of key metrics (e.g., "Revenue")

**2. The Strategic Value of Blue Rose Marts**

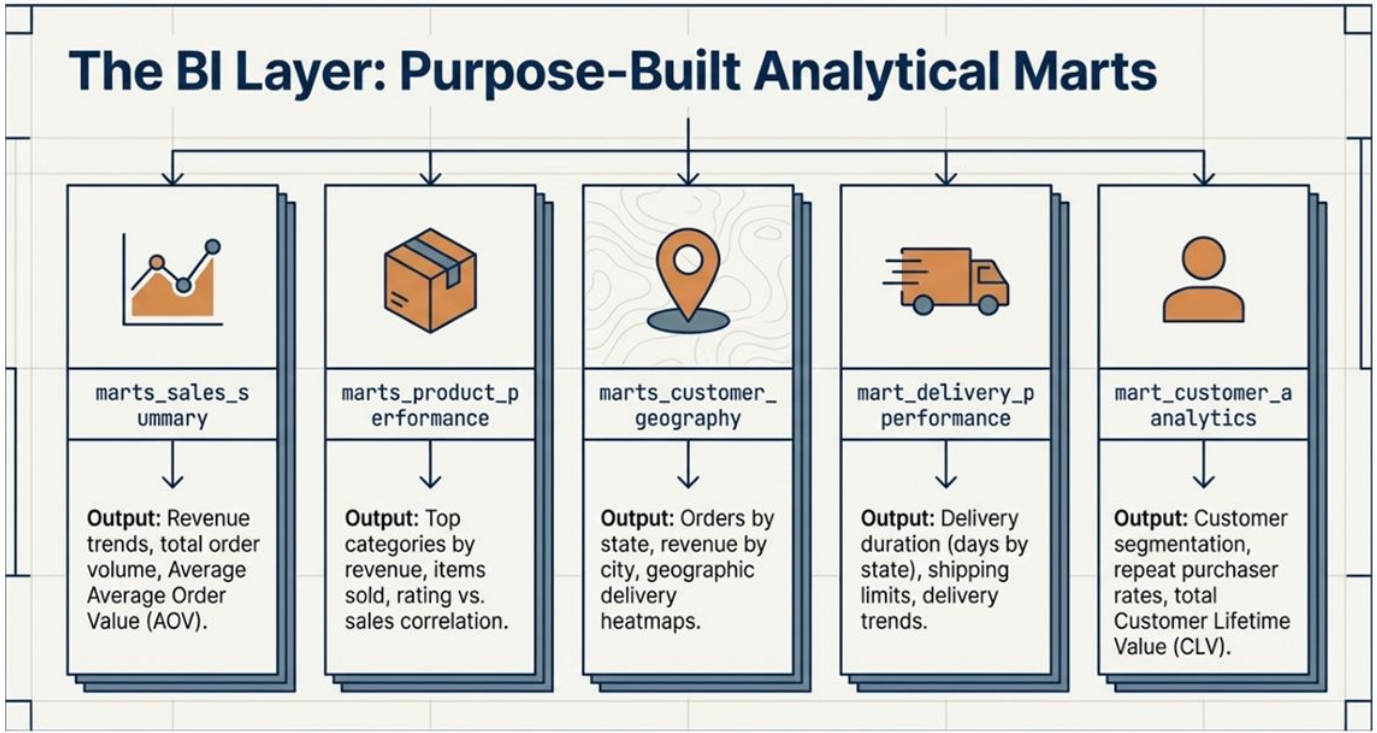
- **Performance**: Pre-aggregated, optimized tables significantly improve dashboard speed (up to 80% faster)
- **Governance**: Role-based access control ensures secure and controlled data usage
- **Consistency**: Standardized KPI definitions create a single source of truth across the organization

**3. Key Outcomes — Blue Rose Marts**

- **Cost Efficiency**: Reduced compute costs by querying smaller, well-modeled datasets instead of raw data
- **Self-Service BI**: Enables analysts to build reports independently without complex SQL joins
- **Reliability**: Ensures high data quality through automated testing in the transformation layer (e.g., dbt tests)



## What Makes up Blue Rose Marts?




**Blue Rose Marts** represent the **business-ready layer** of the Olist data lakehouse, delivering **essential, standardized metrics** while enabling **flexible customization** to support diverse departmental needs.

## How to Leverage Blue Rose Marts?

# Executive Sales Overview *(Jan 2017–Sept 2018)*

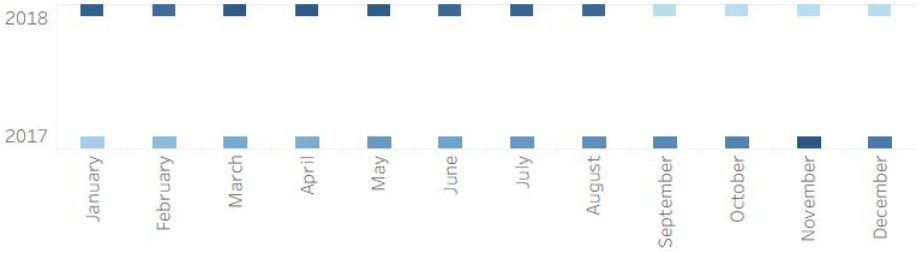
MoM revenue fell 4% in Aug 2018, despite total revenue rising 20% YoY. This demo illustrates the pipeline from Rose Marts to BI, enabling leadership to perform root-cause analysis.

**Future enhancement:**
city/state filter for drill-down insights.

| Total Revenue | Revenue MoM % | Total Order Volume | Avg. Order Value |
|---|---|---|---|
| $8.7M | ▼ 4.1% | 54K | $158 |
| v.s yrs17 (20.1%) | (Aug v.s July 2018 ) | | |

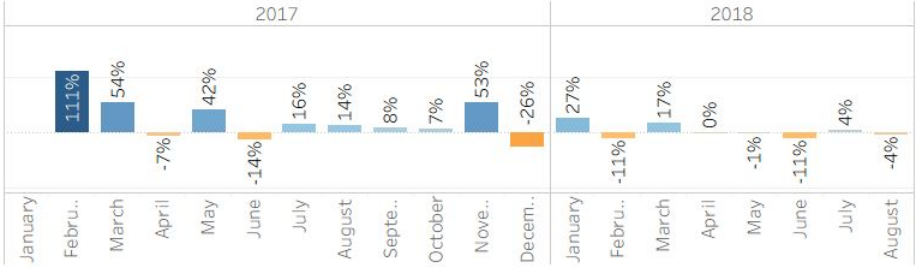
## Revenue Seasonality (Sales Heatmap) updated: 3/22/2026 1:14:47 PM

Dark blue highlights peak sales months.




## Revenue MoM Growth %: updated: 3/22/2026 1:14:47 PM

Spiked in Nov 2017 (Black Friday), and then trended down to -4% by Aug 2018.

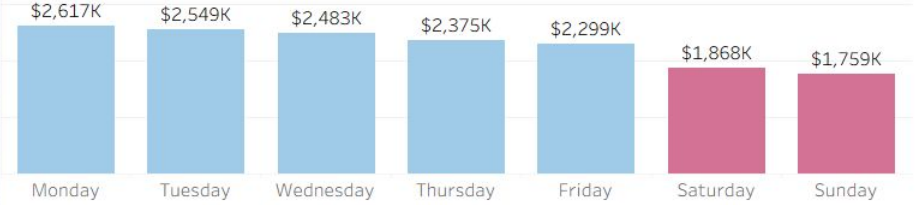



## Average Order Value (AOV) updated: 3/22/2026 1:14:47 PM

Steady around $156, dripped to $147 in Aug, then surged to $224 in Sept 2018.

## Revenue by Day of Week (2018) updated: 3/22/2026 1:14:47 PM

Higher spending observed on weekdays compared to weekends.

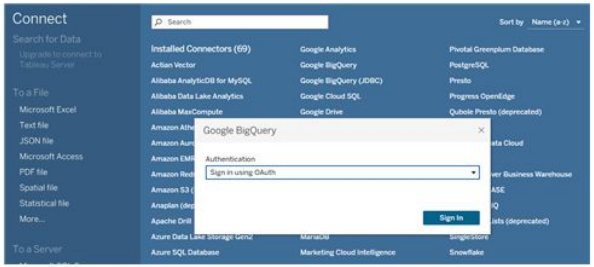


# **Live** connection Blue Rose Marts with full BigQuery data

Switch from the sample data to full BigQuery project...

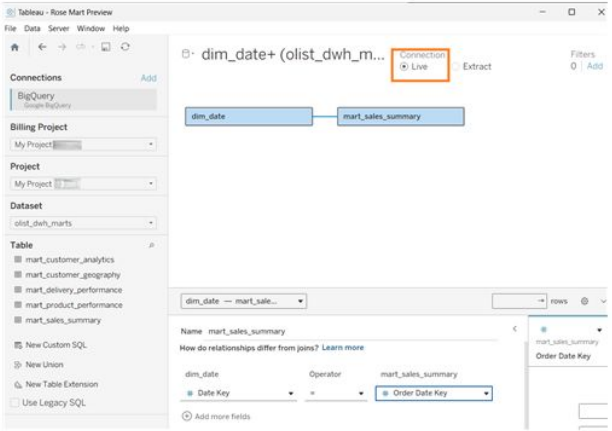
## Step 1: Add the Connection

- In Tableau Desktop, go to the **Data Source** tab or click **Data > New Data Source**.
- Under "To a Server," select **Google BigQuery**.
- Sign in using the Google account that has access to your BigQuery project.




## Step 2: Select Full Data

- Choose **Billing Project**, **Project**, and the **Dataset** where mart layer tables and dim_date table are stored.
- Drag these tables (e.g., Olist dim_date and marts_sales_summary tables) onto the canvas.




## Step 3: Toggle "Live"

- In the top-right corner of the Data Source page, ensure **Live** is selected (instead of **Extract**).

**Note:** With a Live connection, every time you filter or interact with the dashboard, Tableau sends a SQL query to BigQuery. This is great for real-time data but will incur BigQuery query costs based on the amount of data scanned.